# GuitarTap

Tap Tone Analysis Report

March 11, 2026
2:23 PM

**Location:** Plate Classical
**Type:** Material (Plate)
**Notes:** Umik-1
**Frequency Range:** 30.0 Hz – 600.0 Hz
**Microphone:** Umik-1  Gain: 18dB · calibrated (7108913)

## Frequency Spectrum




## Detected Peaks

| Frequency | Magnitude | Note | Q Factor | Role |
|---|---|---|---|---|
| 36.9 Hz | -42.6 dB | D1 | 6.2 | FLC (Diagonal) |
| 67.7 Hz | -47.2 dB | C#2 | 11.5 | Longitudinal (L) |
| 117.4 Hz | -39.5 dB | A#2 | 20.0 | Cross-grain (C) |

**Three-Tap Measurement Process:**

- **1. Longitudinal (L) Tap**
  Hold plate at 22% from one end along the length, near one long edge (not at the width node). Tap center.
- **2. Cross-grain (C) Tap**
  Rotate 90°. Hold plate at 22% from one end along the width, near one short edge (not at the length node). Tap center.
- **3. FLC (Diagonal) Tap**
  Hold plate at the midpoint of one long edge. Tap near the opposite corner (~22% from both the end and the side). Measures shear stiffness.

*The strongest peak from each tap is auto-selected.*

## Plate Properties

**Sample Dimensions**

Length: **557.5 mm** Width: **220.5 mm** Thickness: **4.85 mm**
Mass: **208.0 g** Density: **0.349 g/cm³**

fL: **67.7 Hz** fC: **117.4 Hz** fLC: **36.9 Hz**

Speed of Sound (L): **4219 m/s**
Speed of Sound (C): **1145 m/s**
Young's Modulus (L): **6.21 GPa**
Young's Modulus (C): **0.46 GPa**
GLC (Shear Modulus): **0.372 GPa**

Specific Modulus (L): **17.8** (Fair)
Specific Modulus (C): **1.3** (Very Good)
Radiation Ratio (L): **12.1**
Radiation Ratio (C): **3.3**

Cross/Long Ratio: **0.074**
*typical: 0.04 – 0.08*

Long/Cross Ratio: **13.6**
*typical: 12 – 25*

**Overall Quality:** **Good**

**Gore Target Thickness**

**2.84 mm**

Body: 490 × 390 mm · f_vs = 60 (Classical Top)
GLC (Shear Modulus): 0.372 GPa